

Freight Rate Prediction

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Abstract

This report describes the development of a freight-rate prediction model using the labeled development dataset covering January–October 2025. The main modeling decision was to predict freight rate per mile rather than total posted rate, with a logarithmic target transformation. A tree-ensemble model was progressively regularized and evaluated using chronological validation to simulate future prediction. The final selected configuration was an `ExtraTreesRegressor` with 100 trees and `min_samples_leaf=20`, trained on $\log(\text{posted_rate}/\text{distance})$. Temporal cross-validation produced a mean validation MAE of \$121.13 with a standard deviation of \$2.21 across three future windows. The final model was retrained on all 48,000 labeled rows and generated predictions for all 12,000 rows of the supplied validation set. The provided scorer validated the submission structure and the 31-day December scenario successfully.

1. Problem and Objective

The task is to predict `posted_rate` for unseen freight loads. The labeled development data contains 48,000 rows from January 1 through October 31, 2025. The supplied final validation data contains 12,000 rows from November 1 through December 31, 2025.

The goal was to build a reproducible modeling pipeline that:

1. explores and cleans the development data,
2. uses a validation strategy consistent with the future-dated test set,
3. evaluates multiple modeling hypotheses,
4. avoids target leakage,
5. retrains the selected configuration on all labeled development data, and
6. produces the required `load_id,predicted_rate` submission file.

2. Data Understanding and EDA

2..1 Dataset Structure

The development dataset contains the following fields:

Column	Role / description
load_id	Load identifier; excluded from modeling
pickup	Pickup city
delivery	Delivery city
pickup_lat, pickup_lon	Pickup coordinates
delivery_lat, delivery_lon	Delivery coordinates
distance	Load distance
equipment	Equipment type
weight	Load weight
date	Load date
market_index	Market signal
quote_signal	Quote signal
posted_rate	Target

Table 1: Development dataset fields.

The development target was strongly related to distance. The exploratory analysis found a Pearson correlation of approximately 0.909 and a Spearman correlation of approximately 0.976 between distance and posted rate. The posted-rate distribution was right-skewed, with mean approximately \$2,374, median approximately \$2,031, and a maximum above \$25,000.

Equipment also showed systematic differences in average rate per mile: approximately \$2.12 for Dry Van, \$2.30 for Flatbed, and \$2.38 for Reefer.

2..2 Missing and Invalid Values

Missing values were present in:

- **weight**: 300 training rows,
- **market_index**: 374 training rows.

The validation set had missing values in both **weight** and **market_index** as well.

A separate data-quality inspection found 292 negative weight values in the labeled development data. Because negative freight weight is physically invalid, later experiments converted **weight** < 0 to missing and allowed the preprocessing pipeline to impute it. This correction was evaluated explicitly in EXP-002.

2..3 Future-Dated Validation

A key characteristic of the assessment data is temporal ordering. The labeled development data ends on October 31, while the supplied validation data begins on November 1.

Therefore, a random train/test split was not used as the primary validation strategy. The main internal holdout was:

Train: Jan–Sep 2025 Validation: Oct 2025.

The final November–December validation set was kept untouched during model selection.

3. Modeling Strategy

3..1 Baseline Representation

The initial model used:

- pickup, delivery, equipment as categorical variables;
- numeric features with median imputation;

- categorical features with most-frequent imputation;
- one-hot encoding with `handle_unknown=ignore`;
- no numerical scaling because the initial models were tree ensembles;
- raw date excluded from the model and used for chronological splitting.

The initial baseline was a Random Forest regressor.

3..2 Rate-per-Mile Target

The strongest modeling change was to predict rate per mile instead of total rate:

$$r_i = \frac{\text{posted_rate}_i}{\text{distance}_i}.$$

The final successful formulation further modeled the logarithm:

$$y_i = \log(r_i) = \log\left(\frac{\text{posted_rate}_i}{\text{distance}_i}\right).$$

At prediction time:

$$\hat{r}_i = \exp(\hat{y}_i),$$

followed by:

$$\widehat{\text{posted_rate}}_i = \hat{r}_i \times \text{distance}_i.$$

This formulation separates the basic scale effect of distance from the remaining variation in pricing intensity.

4. Experiment History

The experiments were intentionally incremental so that each major change could be interpreted independently.

Exp.	Change	Oct MAE	Oct RMSE	Oct R^2
001	Random Forest baseline; raw target	\$162.30	\$690.70	0.7958
002	Negative weights $\rightarrow$ missing	\$162.72	\$691.21	0.7955
003	Random Forest, <code>min_samples_leaf=5</code>	\$141.90	\$662.57	0.8121
004	Rate-per-mile target	\$134.67	\$656.39	0.8156
005	Log rate-per-mile target	\$133.67	\$657.74	0.8148
006	Historical target-derived pricing features	\$194.99	\$677.84	0.8034
007	Calendar features	\$205.07	\$672.95	0.8062
008	Explicit pickup-to-delivery route category	\$133.76	\$657.64	0.8149
009	Extra Trees, <code>min_samples_leaf=5</code>	\$130.14	\$658.35	0.8145
010	Extra Trees, <code>min_samples_leaf=10</code>	\$124.92	\$656.41	0.8156

Exp.	Change	Oct MAE	Oct RMSE	Oct R^2
011	Extra Trees, min_samples_leaf=20	\$122.53	\$653.30	0.8173

4.1 What the Experiments Showed

Several observations were important:

- The raw Random Forest baseline had a large training/validation gap.
- Regularization consistently improved future validation as min_samples_leaf increased from 5 to 20.
- Predicting rate per mile improved the validation metrics compared with predicting total rate directly.
- Log-transforming rate per mile gave the lowest October MAE among EXP-001–EXP-011, but the improvement over EXP-004 was small and RMSE/ R^2 moved slightly in the opposite direction.
- Historical target-derived features degraded future performance despite high validation coverage, so they were removed.
- Calendar features degraded future performance and were removed.
- An explicit route categorical feature added negligible value; validation known-route coverage was 99.84%.
- Extra Trees produced lower MAE than the corresponding Random Forest configuration.

5. Error Analysis

The strongest remaining error pattern was concentrated in the extreme upper tail of rate per mile.

For the October holdout, the rate-per-mile distribution had:

$$P_{99} \approx 3.29, \quad P_{99.5} \approx 6.87, \quad P_{99.9} \approx 10.13.$$

The overall EXP-004 error analysis showed:

$$\text{MAE} \approx \$134.67, \quad \text{median absolute error} \approx \$48.56, \quad P_{99}(\text{absolute error}) \approx \$1,728.86.$$

The model also showed systematic underprediction:

$$\text{mean signed error} \approx -\$50.41.$$

High-distance loads had larger absolute errors than short loads. For example, October MAE increased from approximately \$33.90 for loads with distance at most 250 miles to approximately \$247.82 for loads above 2,000 miles.

The extreme high-rate cases were generally not caused by unseen pickup or delivery categories. In the inspected October loads above \$6,000, pickup and delivery categories were already present in training.

These diagnostics motivated the focus on target formulation and regularization rather than adding many more categorical features.

6. Temporal Cross-Validation

After the model family and regularization improved, continued tuning on a single October holdout risked overfitting the selection process to that month. Therefore, EXP-012 used expanding-window temporal cross-validation.

Fold	Training window	Validation window	MAE	R^2
1	Jan–Jun	Jul–Aug	\$122.28	0.8225
2	Jan–Jul	Aug–Sep	\$118.59	0.8290
3	Jan–Sep	Oct	\$122.53	0.8173

Table 3: Chronological cross-validation for the selected configuration.

Across the three folds:

$$\text{Mean MAE} = \$121.13, \quad \text{Std. MAE} = \$2.21,$$

$$\text{Mean RMSE} = \$633.74, \quad \text{Mean } R^2 = 0.8230.$$

The relatively small MAE variation across folds supported using the configuration from EXP-011 for final training.

7. Final Model

The final model was fixed as:

- Model: `ExtraTreesRegressor`
- Number of trees: 100
- `min_samples_leaf`: 20
- Random seed: 42
- Target: `log(posted_rate/distance)`
- Negative weight handling: `weight < 0 → NaN`
- Numeric missing values: median imputation
- Categorical missing values: most-frequent imputation
- Categorical encoding: one-hot, unknown categories ignored

The raw date was not used as a predictive feature in the final model.

The model was then retrained on all 48,000 labeled development rows, covering January 1 through October 31, 2025.

8. Final Validation Predictions

The final model generated predictions for all 12,000 rows in the supplied validation set, covering November 1 through December 31, 2025.

The resulting submission file is:

`validation_predictions.csv`

with exactly the required columns:

load_id,predicted_rate.

Basic output diagnostics were:

- Number of predictions: 12,000
- Minimum predicted rate: \$178.71
- Median predicted rate: \$2,002.81
- Mean predicted rate: \$2,325.31
- Maximum predicted rate: \$6,707.42

The final model artifact was saved as `final_model/model_pipeline.joblib`.

9. December Scenario

The supplied fixed December scenario keeps the load attributes constant and varies only the date. Because the final model intentionally does not use date as a model feature, identical model inputs produce the same prediction for each day.

The generated December prediction was:

\$803.92

for each of the 31 December dates.

This is a consequence of the final feature set rather than an interpolation or manually imposed daily rate.

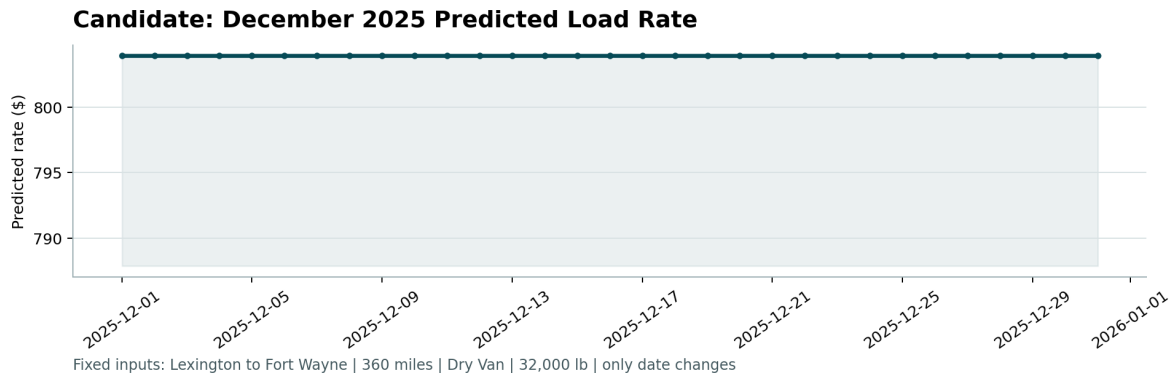


Figure 1: Fixed December 2025 scenario prediction chart generated by the supplied scorer.

10. Submission Verification

The supplied scoring script was run against the final files.

The scorer reported:

Validated 12,000 final predictions.

Validated 31 fixed December predictions.

Created chart: `scorer_results/candidate_december.png`

Final validation metrics are calculated by Spotter after submission.

Therefore, the local submission-format checks passed successfully. The hidden November–December prediction accuracy is not available from the local scorer; it is evaluated by Spotter after submission.

11. Conclusion

The modeling process converged on a relatively simple, reproducible solution.

The most important improvements came from:

1. choosing chronological validation rather than a random split,
2. modeling rate per mile instead of total posted rate,
3. applying a log transformation to rate per mile as a candidate formulation,
4. switching from Random Forest to Extra Trees, and
5. increasing tree regularization with `min_samples_leaf=20`.

Several seemingly plausible additions—historical target-derived features, calendar features, and an explicit route category—did not improve future validation and were therefore excluded from the final pipeline.

The final model was selected using temporal cross-validation and then retrained on all available labeled data. The resulting 12,000-row submission and 31-row December scenario both passed the provided scorer.

Reproducibility

The implementation uses Python and scikit-learn. Experiment scripts, configuration files, metrics, and prediction artifacts are organized by experiment under `experiments/`, with the final pipeline under `final_model/`.

For the assessment submission, the repository should include:

- source code and run instructions,
- `validation_predictions.csv`,
- this report,
- `candidate_december.png`,
- and the short Loom walkthrough.